Similar computations give

$$[\mathcal{L}_{\mathrm{A}}^{\dagger}, \mathcal{L}_{\mathrm{O}}^{\dagger}]\rho = \gamma p \rho_q + \sigma^2 \rho_{qp},$$
$$[\mathcal{L}_{\mathrm{B}}^{\dagger}, \mathcal{L}_{\mathrm{O}}^{\dagger}]\rho = \gamma U'(q) \rho_p.$$

To first order, the discretization map corresponding to scheme ⟦OBA⟧ alters the phase space density at each step as if it were solving the time dependent PDE

$$\rho_t = \hat{\mathcal{L}}^{\dagger} \rho,$$

where

$$\hat{\mathcal{L}}^{\dagger}\rho = \mathcal{L}_{\mathrm{LD}}^{\dagger}\rho + \frac{h}{2}\left[-pU''(q)\rho_p + U'(q)\rho_q + \gamma p \rho_q + \sigma^2 \rho_{qp} + \gamma U'(q)\rho_p\right] + \mathcal{O}(h^2).$$

The validity of this sort of expansion, even for small h, should be considered carefully. Defining

$$\mathcal{L}_{\mathrm{LD}}^{\dagger} = \mathcal{L}_{\mathrm{A}}^{\dagger} + \mathcal{L}_{\mathrm{B}}^{\dagger} + \mathcal{L}_{\mathrm{O}}^{\dagger}$$

as the Kolmogorov operator for Langevin dynamics, the assumption of validity becomes more credible as we find that the operator $\mathcal{L}_{\mathrm{LD}}^{\dagger}$ has a compact resolvent and therefore a discrete spectrum, i.e. simple eigenvalues, as discussed in the previous chapter. It will have one zero eigenvalue corresponding to the invariant density ρ_β and all the remaining eigenvalues lie in the negative half plane. Of course it is not just this operator whose properties are important, but also all the other operators arising in the expansion, and bounding these in relation to $\mathcal{L}_{\mathrm{LD}}^{\dagger}$ is not a simple task (particularly for higher orders).

7.7.3 Perturbation Technique

Suppose we are able to derive a series expansion for $\hat{\mathcal{L}}^{\dagger}$ of the form

$$\hat{\mathcal{L}}^{\dagger} = \mathcal{L}_{\mathrm{LD}}^{\dagger} + h\mathcal{L}_1^{\dagger} + h^2\mathcal{L}_2^{\dagger} + \mathcal{O}(h^3),$$

for some perturbation operators $\mathcal{L}_i^{\dagger}$. We assume that the invariant density has a similar expansion in powers of the stepsize, of the form given in (7.17). Inserting this ansatz into the stationary Fokker-Planck equation

$$\hat{\mathcal{L}}^{\dagger}\hat{\rho} = 0,$$

we obtain

$$\left(\mathcal{L}_{\mathrm{LD}}^{\dagger} + h\mathcal{L}_1^{\dagger} + h^2\mathcal{L}_2^{\dagger} + \mathcal{O}(h^3)\right)\left(\rho_\beta[1 + hf_1 + h^2 f_2 + \mathcal{O}(h^3)]\right) = 0.$$

Combining like powers, we have at zero'th order, the equation

$$\mathcal{L}_{\mathrm{LD}}^{\dagger}\rho_\beta = 0,$$

which we know to hold since Langevin dynamics preserves the canonical ensemble. At first order we obtain

$$\mathcal{L}_{\mathrm{LD}}^{\dagger}(\rho_\beta f_1) + \mathcal{L}_1^{\dagger}\rho_\beta = 0. \tag{7.19}$$

If $\mathcal{L}_1^{\dagger}$ is known, we can calculate its action on ρ_β, giving an inhomogeneity for the above partial differential equation. It will be difficult to solve the perturbation equation (7.19) in general, although its linearity and the fact that $\mathcal{L}_{\mathrm{LD}}^{\dagger}$ (with our assumptions) has a compact resolvent (and therefore a discrete spectrum) together imply that if the perturbation can be bounded, then it should be possible to solve for the correction function. We expect that this solution should be stable with respect to perturbation (and exponentially attracting). This also implies global existence (for all positive time) of the semigroup defining the solution of the time-dependent problem. We do not provide such a rigorous result here but simply plunge ahead, in the best tradition of applied mathematics, to obtain a perturbative expansion; verification of our results must therefore be obtained by numerical experiment, *a posteriori*.

The leading order errors can be computed in simulation by exploiting the specific form of our ansatz for the invariant measure associated to the discretized dynamics. For some observable $\vartheta(\boldsymbol{q},\boldsymbol{p})$, we have that

$$\begin{aligned}
\langle\vartheta\rangle_h &= \int_{\mathcal{D}} \vartheta(\boldsymbol{q},\boldsymbol{p})\hat{\rho}(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega,\\
&= \int_{\mathcal{D}} \vartheta(\boldsymbol{q},\boldsymbol{p})\rho_\beta(\boldsymbol{q},\boldsymbol{p})(1 + hf_1(\boldsymbol{q},\boldsymbol{p}) + h^2 f_2(\boldsymbol{q},\boldsymbol{p}) + \ldots)\mathrm{d}\omega,\\
&= \int_{\mathcal{D}} \vartheta(\boldsymbol{q},\boldsymbol{p})\rho_\beta(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega + h\int_{\mathcal{D}} \vartheta(\boldsymbol{q},\boldsymbol{p})f_1(\boldsymbol{q},\boldsymbol{p})\rho_\beta(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega + \ldots,\\
&= \langle\vartheta\rangle + h\langle\vartheta f_1\rangle + h^2\langle\vartheta f_2\rangle + \ldots.
\end{aligned}$$

Hence by obtaining the correction functions f_i, we are able to apply a modification to our computed expectation in order to recover the true canonical average.